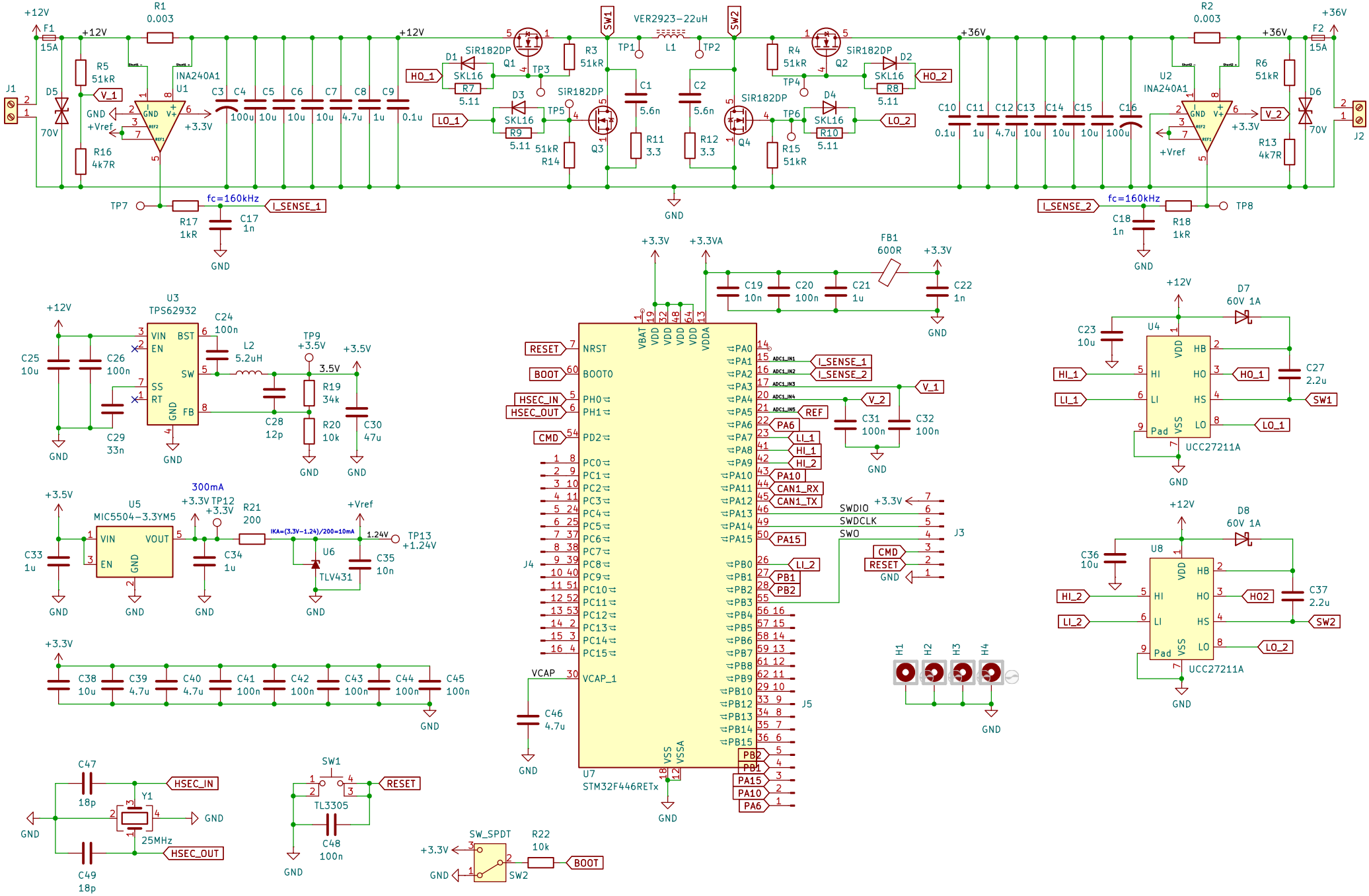

+12V
F1
15A
R1
0.003
INA240A1
U1
R5
51kR
R16
4k7R
V_1
+Vref
+3.3V
C3 100u
C4 10u
C5 10u
C6 10u
C7 4.7u
C8 1u
C9 0.1u
J1
D5
70V
TP7
R17
1kR
fc=160kHz
C17
1n
I_SENSE_1
GND
SW1
SW2
VER2923-22uH
TP1
TP2
L1
D1
SiR182DP
Q1
SKL16
R7
5.11
HO_1
TP3
R3
51kR
C1
5.6n
R11
3.3
LO_1
D3
R9
TP5
R14
Q3
C2
R12
Q4
R15
R4
Q2
D2
R8
HO_2
TP4
TP6
D4
R10
LO_2
+36V
R2
0.003
F2
C10 0.1u
C11 1u
C12 4.7u
C13 10u
C14 10u
C15 10u
C16 100u
U2
R6
51kR
R13
4k7R
D6
J2
I_SENSE_2
TP8
C18
1n
R18
1kR
U3
TPS62932
VIN
EN
SS
RT
BST
SW
GND
FB
C24
100n
L2
5.2uH
C25
10u
C26
100n
C29
33n
C28
12p
TP9
+3.5V
R19
34k
R20
10k
C30
47u
U5
MIC5504-3.3YM5
300mA
TP12
R21
200
IKA=(3.3V-1.24)/200=10mA
1.24V
TP13
U6
TLV431
C33
1u
C34
1u
C35
10n
C38 10u
C39 4.7u
C40 4.7u
C41 100n
C42 100n
C43 100n
C44 100n
C45 100n
C47
18p
HSEC_IN
Y1
25MHz
HSEC_OUT
C49
18p
SW1
TL3305
RESET
C48
100n
SW_SPDT
R22
10k
BOOT
SW2
+3.3VA
FB1
600R
C19
10n
C20
100n
C21
1u
C22
1n
U7
STM32F446RETx
NRST
BOOT0
PH0
PH1
PD2
VBAT
VDD
VDDA
VCAP_1
VSS
VSSA
C46
4.7u
J4
J5
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB12
PB13
PB14
PB15
ADC1_IN1
ADC1_IN2
ADC1_IN3
ADC1_IN4
ADC1_IN5
I_SENSE_1
I_SENSE_2
REF
CAN1_RX
CAN1_TX
SWDIO
SWDCLK
SWO
J3
H1
H2
H3
H4
U4
U8
HI
LI
HO
LO
HS
VDD
VSS
Pad
UCC27211A
D7
D8
60V 1A
C23
10u
C27
2.2u
C36
10u
C37
2.2u
HO2